Adrià Mitjavila Ventura

Bioinformatician

Girona (Spain) amitjavilaventura.github.io amitjavilaventura

Bioinformatician with training in Molecular Genetics and Biotechnology and over 5 years of experience in omics data analysis, mainly transcriptomics and epigenomics.

Employment

Predoctoral researcher, *IJC* & *Institut Germans Trias i Pujol (IGTP)*, Badalona (Spain) *11/2022 - Present*

Analysis and integration of small RNA-seq, RNA-seq data.
Analysis of single-cell and spatial transcriptomics (10x Xenium) data. Design of 10x Xenium custom panel.
Writing of scientific papers and reviews.
Mentoring of undergraduate and master students.
Outreach workshops addressed to high school students.

Bioinformatician *Institut Josep Carreras (IJC)*, Badalona (Spain) *08/2021 - 11/2022*

Analysis and integration of small RNA-seq, RNA-seq, ChIP-seq and other genomics data.
Collaboration with external researchers.

Bioinformatician *Istituto Europeo di Oncologia (IEO)*, Milan (Italy) *07/2019 - 08/2021*

Analysis and integration of RNA-, ChIP- and ATAC-seq data in the context of colorectal cancer.
Comparison of distinct peak callers for ATAC-seq data.
Development of a Snakemake pipeline for ATAC-seq data processing and analysis.
Comparison of sequencing depths for RNA-seq and their impact on differential expression analysis.

Master's thesis student & **Research assistant**, *Instituto de Biomedicina de Sevilla (IBiS)*, Seville (Spain) *09/2017 - 10/2018*

Analysis of MNase-seq data.
Design of PCR primers, performing and analysis of MNase-qPCR data.

Research internship & **Bachelor's thesis student**, *Microbial*, Girona (Spain) *09/2016 - 07/2017*

Analysis of mucosa-associated intestinal microbiota from human biopsies.
Analysis of food and water samples for pathogen detection and food forensics.
Statistical analysis of metagenomics data (with SPSS).
DNA extraction, qPCR, molecular cloning, bacterial cultures.
Design of qPCR kits.

Education

PhD in Bioinformatics, *Universitat Autònoma de Barcelona (UAB)*, Online *2022 - Present*

MSc. Bioinformatics and Biostatistics, *Universitat Oberta de Catalunya (UOC)*, Online *2018 - 2022*

MSc. Molecular Genetics and Biotechnology, *Universidad de Sevilla (US)*, Seville (Spain) *2017 - 2018*

BSc. Biotechnology, *Universitat de Girona (UdG)*, Girona (Spain) *2012 - 2017*

Skills

Languages: Catalan (*native*), Spanish (*bilingual*), English (*IELTS 7.5 - C1*), Italian (*basic*)

Bioinformatics: (small) RNA-/ChIP-/ATAC-/MNase-seq, single-cell RNA-seq, 10x Xenium...

Informatics: R/Bioconductor, R Markdown, Bash, AWK, Python, Snakemake, Conda, Docker, Singularity, HPC (PBS/Slurm)

Publications

Mitjavila-Ventura & Vavouri. **Mechanisms seeding the emergence of new piRNAs**. *Trends in Genetics*. 2026. doi: 10.1016/j.tig.2026.04.004

Casas, Mitjavila-Ventura *et al*. **Genetic polymorphisms lead to major, locus-specific, variation in piRNA production in mouse**. *EMBO Journal*. 2025. DOI: 10.1038/s44318-025-00475-4.

Begley *et al*. **Xrn1 influence on gene transcription results from the combination of general effects on elongating RNA pol II and gene-specific chromatin configuration**. *RNA Biology*. 2021. DOI: 10.1080/15476286.2020.1845504.